4190.428 001 Introduction to Machine Learning

# Course Orientation

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# Course goal

This course aims to learn and understand the idea of representative machine learning algorithms and their basic formulations. Students are asked to take exams and conduct assignments.

# Class Logistics

- Introduction to Machine Learning (2026 Spring)
  - Instructor: Prof. Jaesik Park (jaesik.park@snu.ac.kr)
  - Teaching Assistants: Daneul Kim, Daeun Lee, Jung Hong, Soonho Kim
  - Time and Location: PM 3:30 - 5:00 Tue and Thur @ 302-311-1
  - Office hour: TBA (After class finishes)
  - Course board: eTL bulletin board
    (registered students can access it via http://myetl.snu.ac.kr/ )

# What this class is about?

- “Fundamentals” of machine learning (O)
  - What is the machine learning algorithms?
  - How they can be formulated?
  - How different is it from deep learning?
- “Standard algorithms” for machine learning (O)
  - What are the popular models?
  - How the method can be structured?
  - How to implement and train them?

# What this class is NOT about?

- Advanced course for deep learning algorithms (X)
  - No details will be given about computer vision, computer graphics, natural language processing, speech processing…
- Deep learning programming course (X)
  - No comprehensive labs on Python programming
- Making super-intelligent machines (X)
  - Out of scope ☺

# Pre-requisites

- Background
  - Calculus, probability theory, linear algebra
- Programming skill
  - You will need to demonstrate your idea for the assignments
  - Good ability to code in Python

**GPU environment**

This class does not offer any computing resources, such as GPU clusters. To take this course, please be prepared to conduct the required tasks.

# Topics that we will cover (tentative)

- Linear Classifiers
- Kernel methods
- Support Vector Machines
- Generalization & Regularization
- Nonlinear Classifiers (Perceptron)
- Gradient Descent
- Regression
- Unsupervised Learning (Clustering, K-means, EM, PCA)
- Sequential Models (Markov decision process)
- Reinforcement Learning (LQR, DDP, LQG, Policy Gradient)
- Non-parametric methods
- Overview of recent approaches

# Schedule (tentative)

| Date | Class | Course | Note |
|---|---|---|---|
| 2026/03/03 | 1 | Orientation & Introduction | |
| 2026/03/05 | 2 | Linear classifier | |
| 2026/03/10 | 3 | Feature representation | |
| 2026/03/12 | 4 | Kernel methods | [2026/03/13] Assignment #1 announcement |
| 2026/03/17 | 5 | Support vector machines | |
| 2026/03/19 | 6 | Support vector machines | |
| 2026/03/24 | 7 | Support vector machines | |
| 2026/03/26 | 8 | Generalization and regularization | [2026/03/27] Assignment #2 announcement |
| 2026/03/31 | 9 | Generalization and regularization | |
| 2026/04/02 | 10 | Perceptron | [2026/04/03] Assignment #1 deadline |
| 2026/04/07 | 11 | Perceptron | |
| 2026/04/09 | 12 | Perceptron | |
| 2026/04/14 | 13 | Logistic regression | |
| 2026/04/16 | 14 | Generative learning | [2026/04/17] Assignment #2 deadline |
| 2026/04/21 | 15 | Midterm Exam | |
| 2026/04/23 | 16 | Exam Explanation (Optional) | |
| 2026/04/28 | 17 | Gradient descent | |
| 2026/04/30 | 18 | Regression | |
| 2026/05/05 | - | | Children's day |
| 2026/05/07 | 19 | K-menas Clustering | [2026/05/08] Assignment #3 announcement |
| 2026/05/12 | 20 | Variational Auto-encoder | |
| 2026/05/14 | 21 | Principal Component Analysis | |
| 2026/05/19 | 22 | Sequential models | |
| 2026/05/21 | 23 | Sequential models | [2026/05/22] Assignment #4 announcement |
| 2026/05/26 | 24 | Reinforcement learning | |
| 2026/05/28 | 25 | Reinforcement learning | [2026/05/29] Assignment #3 deadline |
| 2026/06/02 | 26 | Reinforcement learning | Video Lectures (ICRA/CVPR) |
| 2026/06/04 | 27 | Non-parametric methods | Video Lectures (ICRA/CVPR) |
| 2026/06/09 | 28 | Non-parametric methods | |
| 2026/06/11 | 29 | Non-parametric methods | [2026/06/12] Assignment #4 deadline |
| 2026/06/16 | 30 | Final Exam | |
| 2026/06/18 | 31 | Exam Explanation (Optional) | Finishing course (Thur, June 18.) |

# Grading (100 + ? pts)

- Assignments (5 x 4 = 20pts)
  - Submitting reports or programming assignments
- Mid-term examination (30pts)
- Final examination (40pts)
- Attendance (10pts)
  - Electronic attendance check + random physical checking
  - If you are late 3 times, you get -1 pts
  - If you are absent, -1 pts (let TAs know if you have a reason)
- Bonus points (?pts)
  - If you want to get A+ from this course
- Absolute scorings
- Note that this plan is subject to change...

# Grading

- Programming platform
  - Python
  - PyTorch (strongly recommend)
- DO NOT COPY your friends' work

**Academic integrity**

Make sure to acknowledge the SNU academic integrity. Violating the academic integrity means the automatic failure (F) in this class and severe penalty from university with NO exception.

# Enrollment

- Currently, the max limit of enrollment is set to 130, but it will can be adjusted.
- Most of students are from CSE and GSAI.

# Be polite and respect others

- Professor and TAs
  - We will try to help
  - Please use polite words
- Please respect others
  - Your friend

https://kidshelpline.com.au/teens/issues/all-about-respect

# QnA

- Any questions?